

KEY CONCEPTS IN CLINICAL EPIDEMIOLOGY**The use of the E-value for sensitivity analysis**William T. Chung^a, Kevin C. Chung^{b,*}^a*Clinical Research Assistant, Section of Plastic Surgery, Department of Surgery, University of Michigan Hospital, Ann Arbor, MI, USA*^b*Professor of Surgery, Section of Plastic Surgery, University of Michigan Medical School, Ann Arbor, MI, USA*

Accepted 26 September 2023; Published online 1 October 2023

Abstract

Observational research designs enable clinicians to investigate topics for which randomized-controlled trials may be difficult to conduct. However, the lack of randomization in observational studies increases the likelihood of confounders introducing bias to study results. Analytical methods such as propensity score matching and regression analysis are employed to reduce the effects of such confounding, mainly by determining characteristics of patient groups and adjusting for measured confounders. Sensitivity analyses are subsequently applied to elucidate the extent to which study results could still be affected by unmeasured confounding. The E-value is one such approach. By presenting a value that quantifies the strength of unmeasured confounding necessary to negate the observed results, the E-value is a useful heuristic concept for assessing the robustness of observational studies. This article provides an introductory overview of how the E-value can be evaluated and presented in clinical research studies. © 2023 Elsevier Inc. All rights reserved.

Keywords: Observational research; Confounding; Sensitivity analysis; E-value

1. Introduction

The practice of evidence-based medicine considers randomized-controlled trials (RCTs) to be the highest quality of research available to clinicians. In a well implemented RCT, randomization ensures that the groups being compared are “equal” in their characteristics by attributing any differences in outcomes to the intervention of interest. However, RCTs are not always feasible. Investigations of population health research may require decades of follow-up time, or pose issues of generalizability when selecting populations more amenable to RCT conditions (e.g., patients at university hospitals) [1]. Surgery is another field where RCTs are difficult to implement. Surgical treatment often cannot be blinded, and the randomization of treatments themselves may present ethical dilemmas [2]. In such cases, observational studies are an effective alternative [3,4].

In the absence of randomization, patient groups in observational research are susceptible to confounding. “Confounders” are defined as risk factors for the studied outcome that are associated with the treatment or exposure, but not influenced by treatment [5]. Consider the following

hypothetical example: when comparing the postoperative complications of different surgical interventions for wrist fractures, the comparator groups are found to have unequal distribution in age, with one treatment group having significantly more older patients than the other. The influence of age on choice of surgical procedure, as well as its effect on complication rates (independent of choice of surgical procedure), become important considerations. Unless these effects of age are accounted for in the analysis, one can expect the comparison of different surgical treatments to be biased by this confounder.

A number of analytical methods exist for bringing observational research more in line with the randomized comparison groups of RCTs; chief among them propensity score methods and regression analysis [3]. Although beyond the scope of this article, these methods seek to assess the characteristics of patient groups and either “match” patients or introduce statistical adjustments to minimize the effect of confounders. However, such adjustments can only eliminate the effects of measured confounding. Sensitivity analyses are employed to quantify the degree to which results may still be affected by unmeasured confounders [6]. The focus of this concepts article, the E-value, is a novel sensitivity analysis that contextualizes the results of observational research in a simple, accessible way.

* Corresponding author. Comprehensive Hand Center, The University of Michigan Health System, 1500 E. Medical Center Drive, 2130 Taubman Center, SPC 5340, Ann Arbor, MI 48109-5340, USA. Tel.: 734-936-5885; Fax: 734-763-5354.

E-mail address: kecchung@med.umich.edu (K.C. Chung).

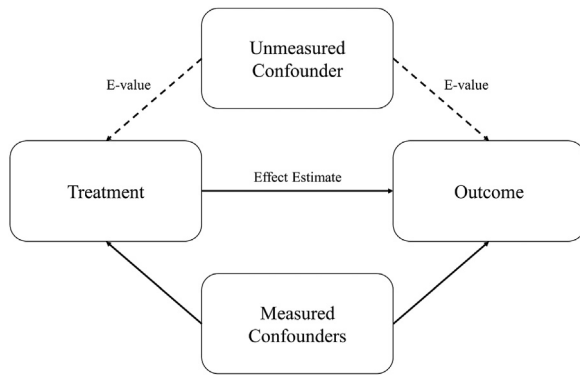


Fig. 1. The E-value quantifies the strength of association that an unmeasured confounder would have to have with both treatment and outcome to nullify the observed treatment-outcome association.

2. Definition

Introduced by Vanderweele et al., [7] the E-value represents the extent to which an unmeasured confounder would have to be associated with both treatment and outcome to nullify the observed treatment-outcome association (Fig. 1). E-values are measured on the ratio scale. Greater E-values indicate stronger treatment-outcome associations, as unmeasured confounders would have to have correspondingly large risk ratios (RRs) to negate the results. The lowest an E-value can be is one, in which case no unmeasured confounding is necessary to nullify the observed association.

For any given RR reported in a study, two E-values can be calculated: one for the observed outcome, and one for the 95% confidence interval (CI) of the outcome, with the latter evaluated for the limit of the CI closest to the null [7]. The E-value for the observed outcome is calculated with the following equation.

$$E\text{ value} = RR + \sqrt{RR \times (RR - 1)}$$

For RRs less than 1, RR in the above equation is replaced with its inverse, $1/RR$. The E-value of the 95% CI is calculated using the equations in Table 1. Including this second measure ensures that statistical uncertainty is accounted for when reporting these values. Vanderweele et al. [7] provide additional guidance on how the equation above and those in Table 1 can be adapted to calculate E-values for other effect estimates, including odds ratios and hazard ratios (Supplement).

3. Strengths and limitations

As the equations demonstrate, a clear strength of the E-value is that its calculation relies on existing effect

estimates and CIs and does not require any assumptions from the investigators [8]. However, this same quality has been criticized for lending the E-value an almost “linear” relationship with its corresponding effect estimate [9]. Vanderweele et al. [10] have countered these concerns by presenting the example of *P* values and CIs; either measure can be derived from the other, but both are usually reported as they illustrate different qualities of the study results. Along with its 95% CI counterpart, the E-value adds another such dimension to the presentation of results by providing a simple metric for assessing the susceptibility of findings to confounding. In contrast to more complex sensitivity analyses, this heuristic method may be readily applied to current observational research to assess levels of confounding bias. To this end, reviews implementing the E-value have been conducted for observational surgical research, revealing the extent to which such studies are likely affected by confounding and giving impetus for more robust study design [11,12].

Context is crucial when evaluating E-values. By itself, an E-value of 5 holds no meaning. However, when compared to known risk factors with RRs in the 6–7 range, the smaller E-value of 5 indicates outcomes susceptible to confounding [8]. Conversely, an E-value of 2 may appear small at face value. Yet if known risk factors have RRs around 1.5, the comparably greater E-value of 2 may suggest a robust, causal link between treatment and outcome. The E-value should not be considered a substitute for the application of established statistical adjustments such as propensity score methods or multivariable analysis, or for more in-depth sensitivity analyses [10]. The E-value is a method to be used *in addition to* these. Studies that make every attempt to minimize confounding, and which then employ E-values to further contextualize the strength (or weakness) of their outcomes will demonstrate greater robustness than those that rely solely on E-values. The strength of the E-value is in its simplicity, which allows researchers to quickly assess weaknesses. Employing more rigorous analyses will provide finer granularity on the confounding biases.

4. Example: use of the E-value in practice

In a 2020 study published in *Science Advances*, Wu et al. [13] examined the relationship between COVID-19 mortality and air pollution across the United States. After adjusting for 20 county-level confounders, Wu et al. determined a statistically significant mortality rate ratio (MRR) of 1.11 (95% CI, 1.06–1.17), indicating an 11% increase in

Table 1. Equations for E-values of 95% confidence intervals

RR > 1	If LL ≤ 1, then E-value = 1
	If LL > 1, then E-value = $LL + \sqrt{LL \times (LL - 1)}$
RR < 1	If UL ≥ 1, then E-value = 1
	If UL < 1, then $UL^* = 1/UL$ and E-value = $UL^* + \sqrt{UL^* \times (UL^* - 1)}$

Where RR = risk ratio, LL = lower limit of 95% confidence interval, and UL = upper limit.

mortality per 1 $\mu\text{g}/\text{m}^3$ increase in long-term average particulate matter 2.5 (particles 2.5 μm or less in diameter, or $\text{PM}_{2.5}$). Further, an E-value for the effect estimate was calculated to provide a measure of robustness against unmeasured confounders. Using the equation described previously, an MRR of 1.11 produces an E-value of 1.46. The equations in Table 1 can also be used to calculate the E-value for the limit of the CI closest to null. Using the lower limit of 1.06, this gives an E-value of 1.31.

These E-values are best presented within the context of known confounders, described by Wu et al. [13] as days since the first COVID case reported, weather variables, number of hospital beds, behavioral risk factors, and longitude/latitude. RRs for these confounders' associations with MRR ranged from 1.00 to 1.15, all at levels well below the E-values for the effect estimate and the CI. These numbers, on top of the statistical adjustments made with the 20 identified confounders, led Wu et al. to suggest a robustness in the treatment-outcome association that could only be explained away by confounders with RRs greater than those observed. In the absence of rigorous analytical adjustments or the contextualization of the E-value to measured confounders, the values of 1.46 or 1.31 mean very little. The above is a good example of how E-values can be used effectively in observational research to improve interpretations of robustness to confounding bias.

5. Pointers

- The E-value provides a value, measured on the ratio scale, of how strongly an unmeasured confounder must be associated with both treatment and outcome to negate the observed treatment-outcome association.
- The E-value does not require any assumptions to be calculated; only the effect estimate and 95% CI limits are needed for calculation.
- Context is important. E-values are meaningless without comparisons to the RRs of known confounders in relation to the outcome.
- E-values are valuable when used in addition to analytical methods for minimizing confounding such as matching or multivariable adjustment. They should not be considered a substitute for claiming study robustness.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

One or more of the authors (K.C.C.) receives funding from the National Institutes of Health, book royalties from

Wolters Kluwer and Elsevier, and a research grant from So-nex to study carpal tunnel outcomes.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2023.09.014>.

References

- [1] Sanson-Fisher RW, Bonevski B, Green LW, D'Este C. Limitations of the randomized controlled trial in evaluating population-based health interventions. *Am J Prev Med* 2007;33(2):155–61.
- [2] Hoppe DJ, Schemitsch EH, Morshed S, Tornetta PI, Bhandari M. Hierarchy of evidence: where observational studies fit in and why we need them. *J Bone Joint Surg Am* 2009;91(Supplement_3):2.
- [3] D'Agostino RB. Estimating treatment effects using observational data. *JAMA* 2007;297:314–6.
- [4] Concato J, Shah N, Horwitz RI. Randomized, controlled trials, observational studies, and the hierarchy of research designs. *N Engl J Med* 2000;342:1887–92.
- [5] Jager KJ, Zoccali C, MacLeod A, Dekker FW. Confounding: what it is and how to deal with it. *Kidney Int* 2008;73(3):256–60.
- [6] Rosenbaum PR. Sensitivity analysis in observational studies. In: *Encyclopedia of statistics in behavioral science*. Hoboken, NJ: John Wiley & Sons, Ltd; 2005.
- [7] VanderWeele TJ, Ding P. Sensitivity analysis in observational research: introducing the E-value. *Ann Intern Med* 2017;167:268–74.
- [8] Haneuse S, VanderWeele TJ, Arterburn D. Using the E-value to assess the potential effect of unmeasured confounding in observational studies. *JAMA* 2019;321:602–3.
- [9] Ioannidis JPA, Tan YJ, Blum MR. Limitations and misinterpretations of E-values for sensitivity analyses of observational studies. *Ann Intern Med* 2019;170:108–11.
- [10] VanderWeele TJ, Mathur MB, Ding P. Correcting misinterpretations of the E-value. *Ann Intern Med* 2019;170:131–2.
- [11] Baxter NB, Kocheril AP, Chung KC. Application of the E-value to assess bias in observational research in plastic surgery. *Plast Reconstr Surg* 2022;150:1151.
- [12] Wang CW, Chung WT, Baxter NB, Chung KC. Are observational studies on distal radius fracture treatment robust? An E-value approach to analysis. *Clin Orthop Relat Res* 2023;481(6):1174.
- [13] Wu X, Nethery RC, Sabath MB, Braun D, Dominici F. Air pollution and COVID-19 mortality in the United States: strengths and limitations of an ecological regression analysis. *Sci Adv* 2020;6(45):eabd4049.

Further Reading

- [1] Haneuse S, VanderWeele TJ, Arterburn D. Using the E-value to assess the potential effect of unmeasured confounding in observational studies. *JAMA* 2019;321:602–3.
- [2] VanderWeele TJ, Ding P. Sensitivity analysis in observational research: introducing the E-value. *Ann Intern Med* 2017;167:268–74.
- [3] Ioannidis JPA, Tan YJ, Blum MR. Limitations and misinterpretations of E-values for sensitivity analyses of observational studies. *Ann Intern Med* 2019;170:108–11.
- [4] VanderWeele TJ, Mathur MB, Ding P. Correcting misinterpretations of the E-value. *Ann Intern Med* 2019;170:131–2.